

'Rente mit 63'

Labour Force Participation among Eligible Older Workers in Germany

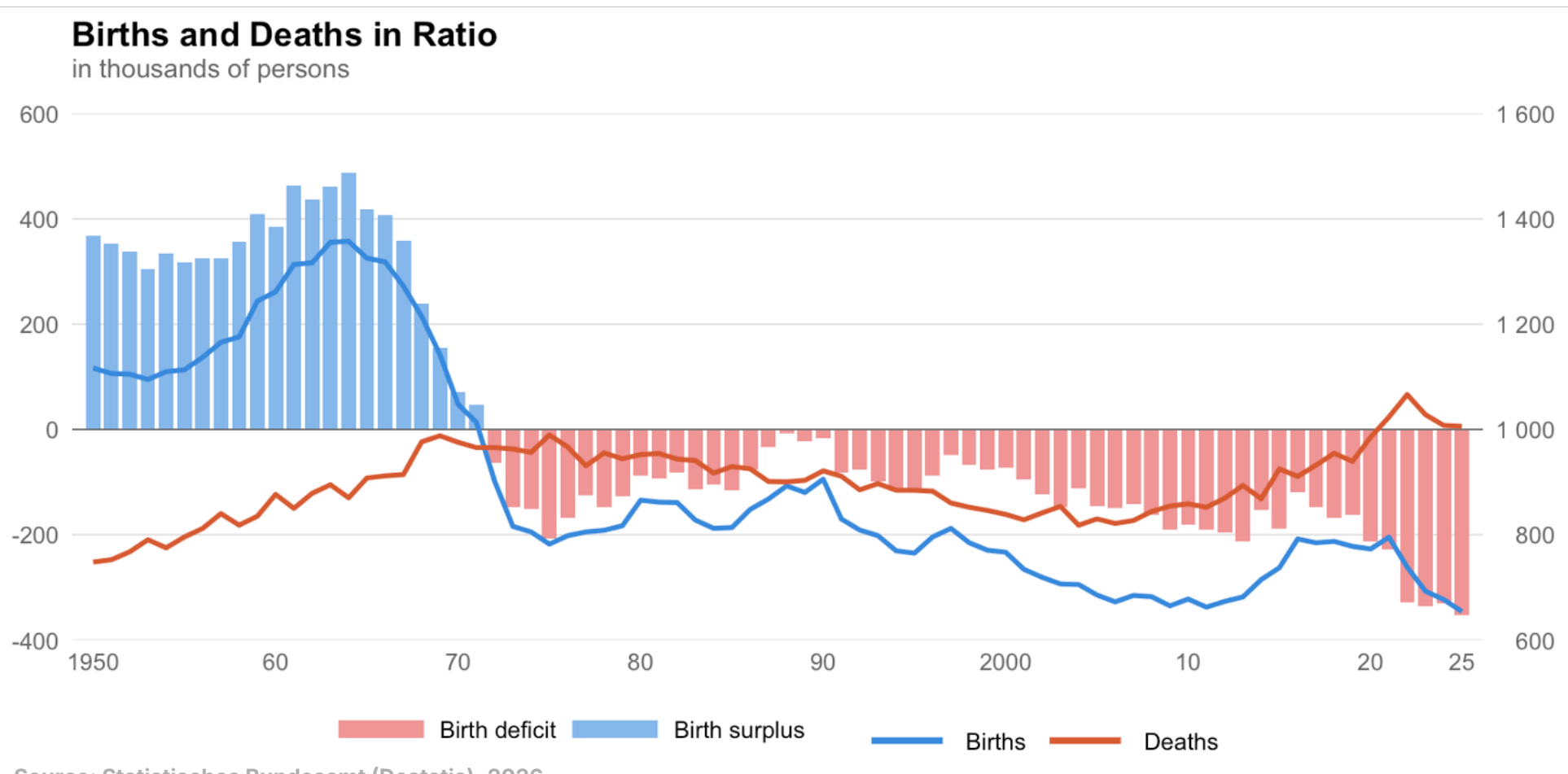
SPS Pareto Seminar - Research Projects in Empirical Economics

Supervisor: Şimal Köseoğlu

Anton Wilhelmi | Konstantin Böhm

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Germany Is Shrinking

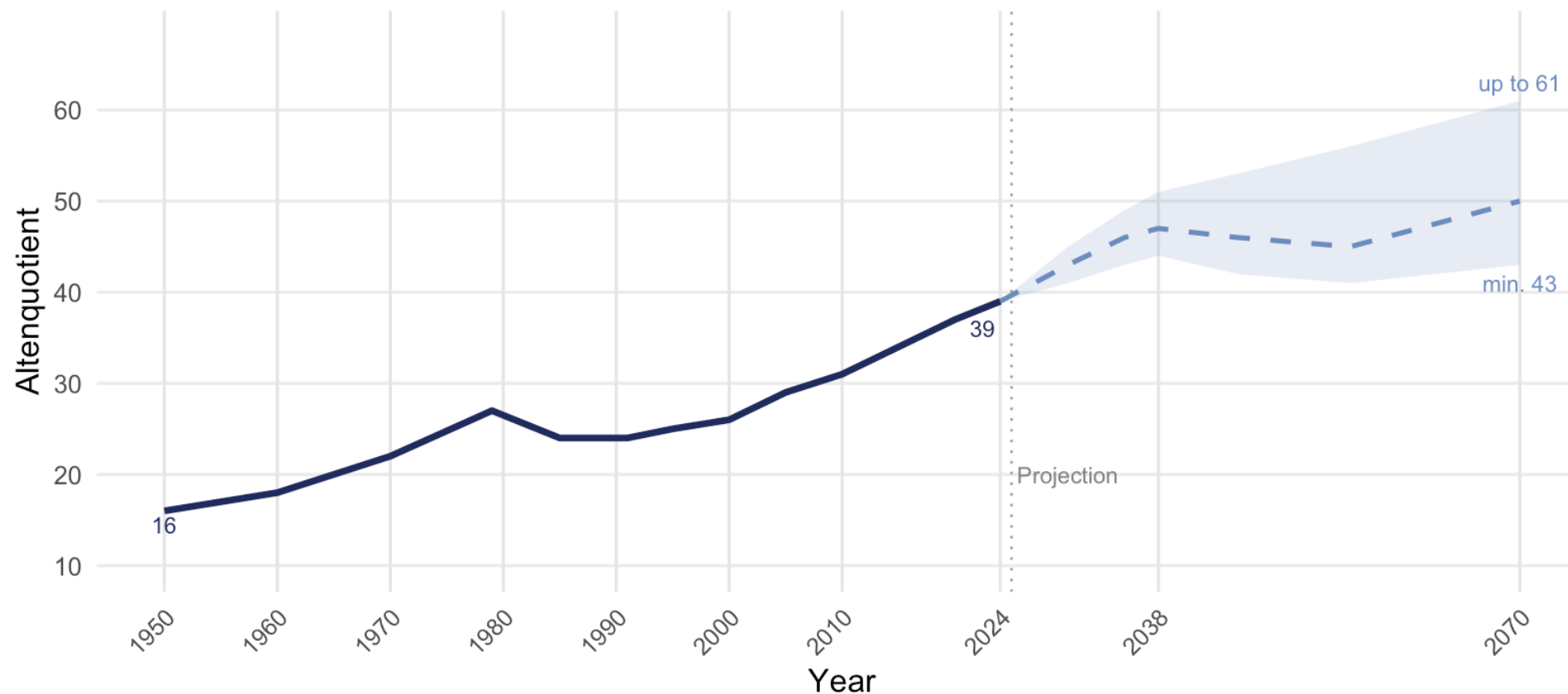


Source: Statistisches Bundesamt (Destatis), 2026

The Demographic Burden Is Growing Fast

Old-Age Dependency Ratio in Germany, 1950 to 2070

Persons aged 65+ per 100 persons aged 20-64 | Shaded band: projection range (favourable to unfavourable scenario)



Source: Destatis, 16. koordinierte Bevölkerungsvorausberechnung (2025). Historical values: Destatis (2024).

How Germany's Pension System Works

- **Pay-as-you-go**
 - *Current workers finance current retirees*

- **Early retirement penalty**
 - *Permanent deduction of 0.3% per month for early retirement*

- **Contribution years as eligibility**
 - *45 qualifying years enable access to early retirement provisions*

Reform: Rente mit 63

- **Implemented**

- *July 2014*

- **Eligibility**

- *45+ contribution years*

- **Retirement age**

- *reduced to 63, no deductions*

Birth Year	Earliest Retirement Age
before 1953	63 + 0 months
1953	63 + 2 months
1954	63 + 4 months
1955	63 + 6 months
1956	63 + 8 months
...	...
1964+	67

Source: Deutsche Rentenversicherung Bund

MAIN RESEARCH QUESTION

Did the introduction of Rente mit 63 significantly reduce labour force participation among eligible workers?

RQ1 · REGIONAL HETEROGENEITY

East vs. West Germany

Does the effect vary between East and West Germany?

RQ2 · SOCIOECONOMIC HETEROGENEITY

RQ2a: Income

Does the effect differ across income groups?

RQ2b: Education

Does the effect differ across education groups?

Difference-in-Differences Framework

Main Regression

$$LFP_{it} = \alpha_i + \beta_1 Post_t + \beta_2 (Treat_i \times Post_t) + \epsilon_{it}$$

RQ1 · Regional Heterogeneity

$$LFP_{it} = \alpha_i + \beta_1 Post_t + \beta_2 (Treat_i \times Post_t) + \beta_3 (Treat_i \times Post_t \times East_i) + \epsilon_{it}$$

RQ2a · Income Heterogeneity

$$LFP_{it} = \alpha_i + \beta_1 Post_t + \beta_2 (Treat_i \times Post_t) + \sum_{k=2}^5 \beta_k (Treat_i \times Post_t \times incq_k) + \epsilon_{it}$$

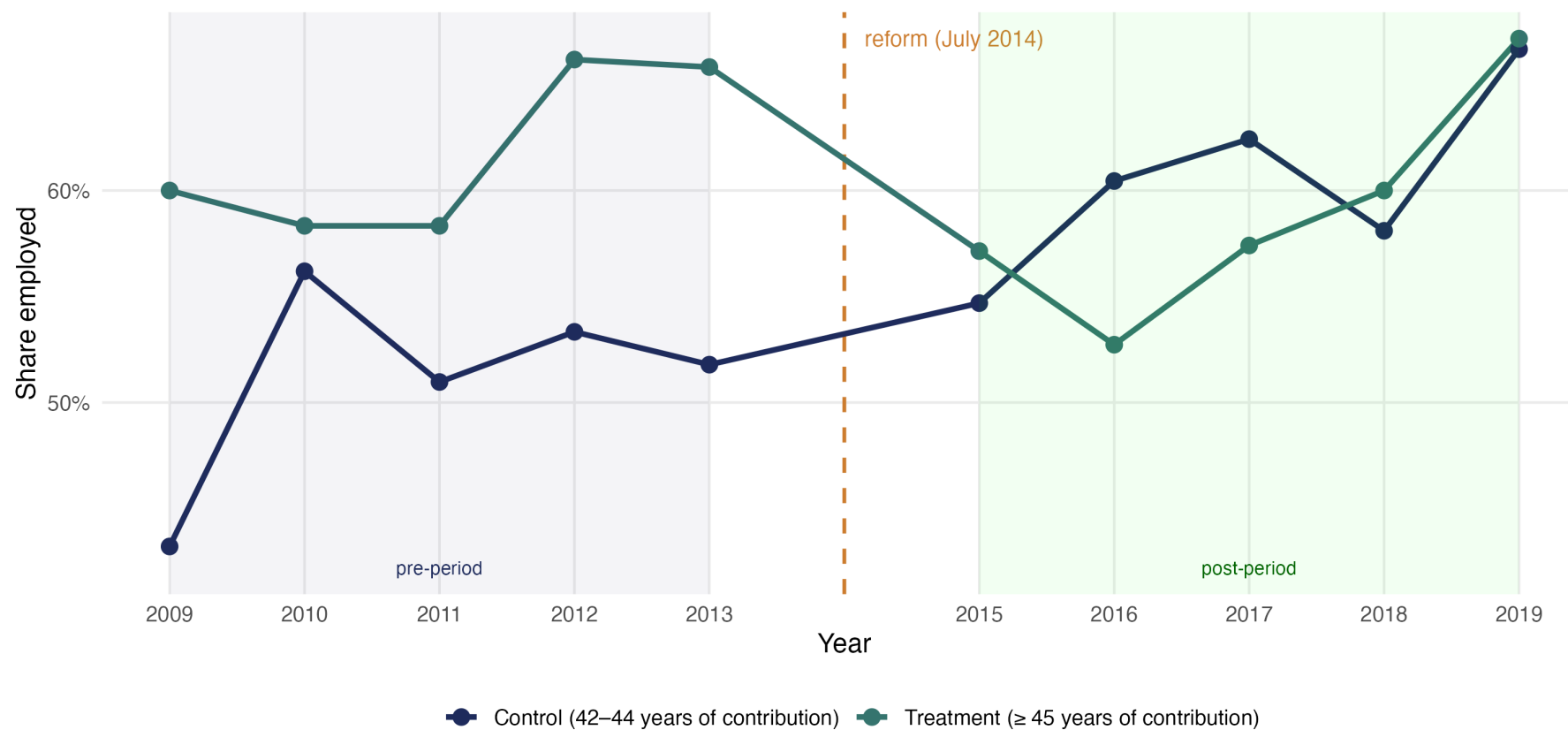
RQ2b · Educational Heterogeneity

$$LFP_{it} = \alpha_i + \beta_1 Post_t + \beta_2 (Treat_i \times Post_t) + \beta_3 (Treat_i \times Post_t \times HighEduc_i) + \epsilon_{it}$$

Control Group Selection

Employed: treatment- vs. controlgroup (2009–2019)

Control: 42–44 contribution years | 2014 excluded | Parallel Trends: $p = 0.097$



Source: SOEP-Data

Main Regression	
<i>Variables</i>	
Post	−0.342*** (0.057)
Treat × Post	−0.219*** (0.050)
Observations	1,674
Individual FE	Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; SOEP-Data

Insignificant Heterogeneity

	RQ1: East/West	RQ2b: Education
<i>Variables</i>		
Post	−0.347*** (0.056)	−0.342*** (0.057)
Treat × Post	−0.173*** (0.052)	−0.212*** (0.058)
Treat × Post × East	−0.172 (0.111)	
Treat × Post × High Educ		−0.016 (0.094)
Observations	1,674	1,674
Individual FE	Yes	Yes

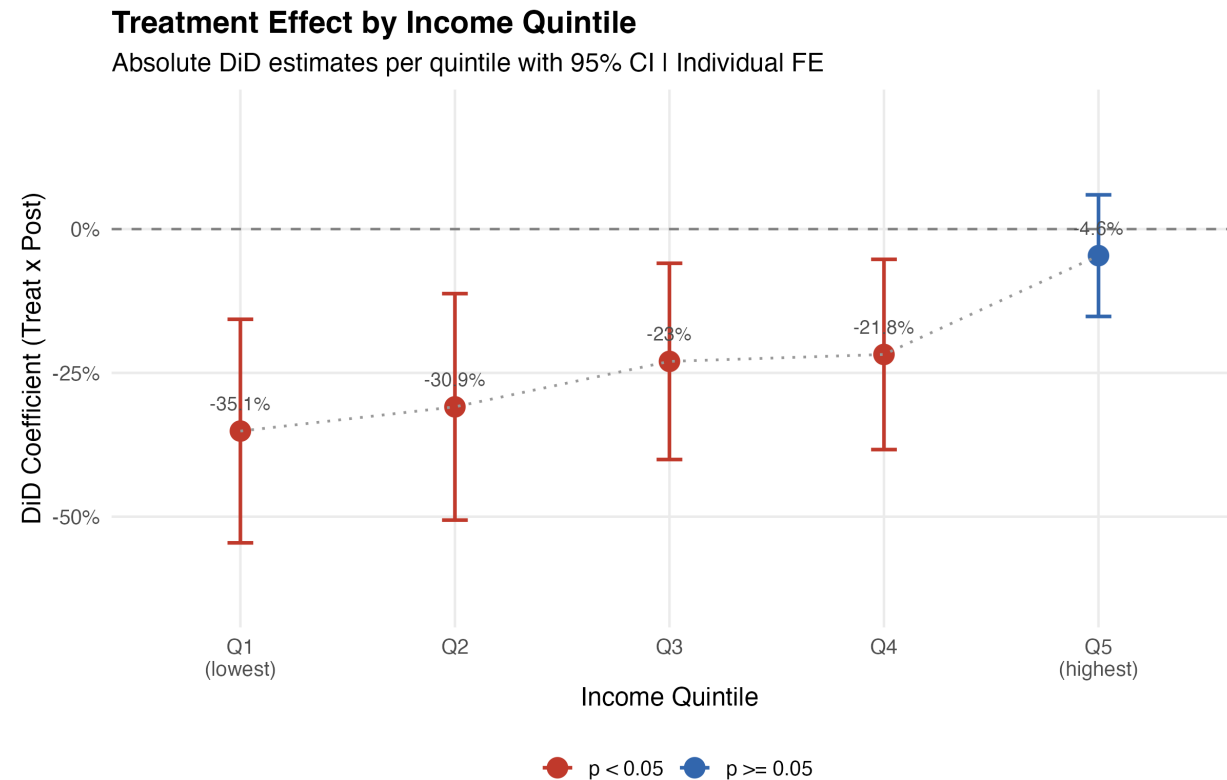
* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; SOEP-Data

Significant Heterogeneity

RQ2a: Income Gradient

Variables	
Post	-0.337*** (0.057)
Treat × Post (Q1 Ref.)	-0.351*** (0.099)
Treat × Post × Q2	+0.042 (0.114)
Treat × Post × Q3	+0.121 (0.123)
Treat × Post × Q4	+0.133 (0.124)
Treat × Post × Q5	+0.305** (0.110)
Observations	1,674
Individual FE	Yes

* $p < 0.05$, ** $p < 0.01$, *** $p < 0.001$; SOEP-Data



Robustness: Alternative Control Groups

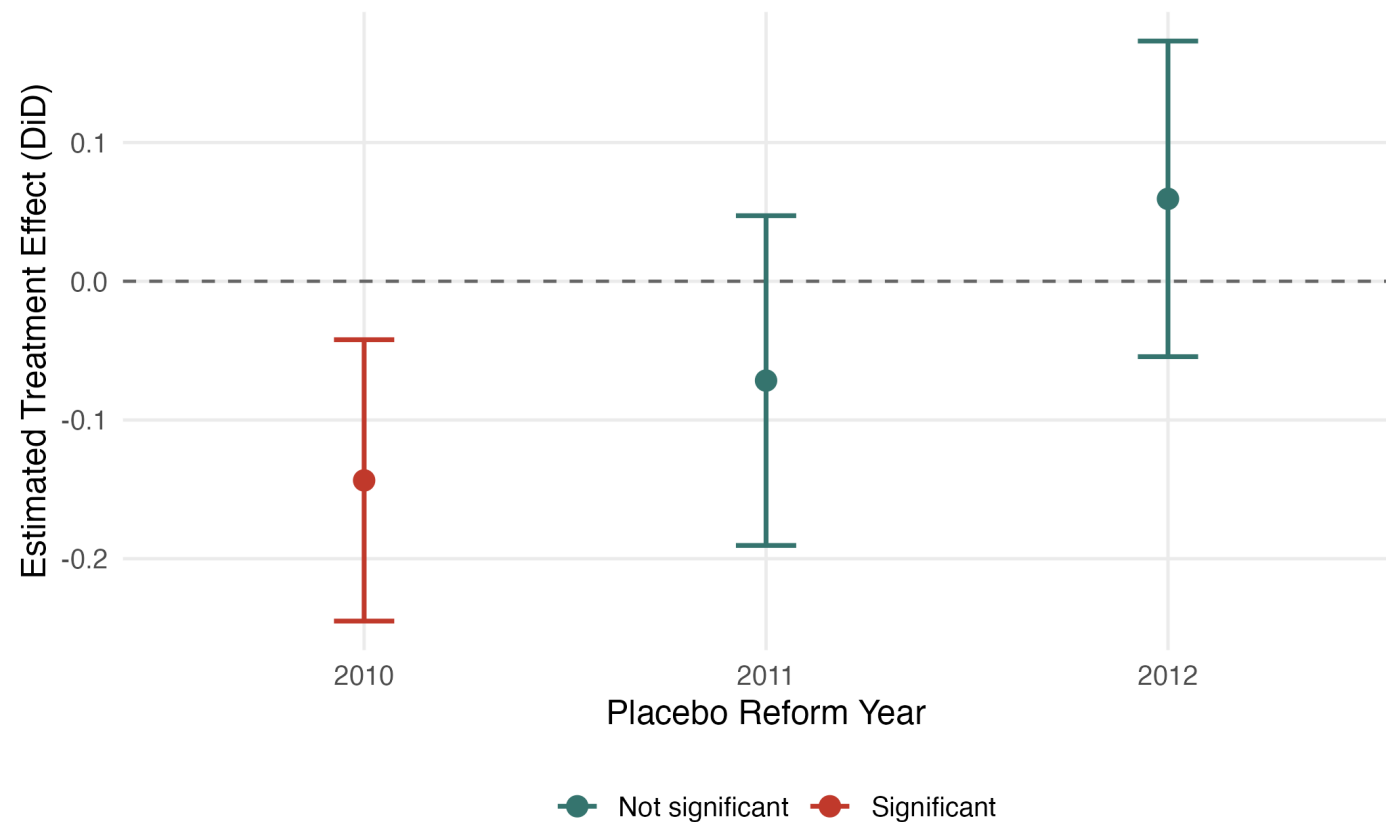
	35–44	40–44	41–44	42–44*
<i>Variables</i>				
Treat	0.239*** (0.038)	0.170*** (0.040)	0.139*** (0.041)	0.110* (0.043)
Post	0.136*** (0.018)	0.103*** (0.025)	0.108*** (0.028)	0.096** (0.032)
Treat × Post	−0.170*** (0.053)	−0.137* (0.055)	−0.142* (0.056)	−0.130* (0.058)
N (Total)	6,553	3,438	2,731	2,068
Ratio (C:T)	10.5:1	5.0:1	3.8:1	2.6:1

* p < 0.05, ** p < 0.01, *** p < 0.001 ; SOEP-Data

Robustness: Placebo Tests

Placebo Test – Estimated DiD Effects

Fake reform years (pre-2014 period only)



Source: SOEP-Data

Robustness: Donut Design

	Full Sample	Donut Sample
<i>Variables</i>		
Post	−0.342*** (0.057)	−0.339*** (0.077)
Treat × Post	−0.219*** (0.050)	−0.194** (0.073)
Observations	1,674	1,232
Individual FE	Yes	Yes

* p < 0.05, ** p < 0.01, *** p < 0.001 ; SOEP-Data

Robustness: Potential Confounder Flexi-Rente

	Baseline	Flexi-Dummy	2015–16 only
<i>Variables</i>			
Post	−0.342*** (0.057)	−0.396*** (0.057)	−0.410*** (0.064)
Treat × Post	−0.219*** (0.050)	−0.114* (0.052)	−0.088 (0.079)
Post-2017 (Flexirente)		−0.342*** (0.045)	
Observations	1,674	1,674	1,072
Individual FE	Yes	Yes	Yes
Post-Period	2015–19	2015–19	2015–16

* p < 0.05, ** p < 0.01, *** p < 0.001 ; SOEP-Data

- **Men only**
 - *Women rarely reach 45 years of contribution*
- **Data**
 - *Small treatment group*
 - *Approximated contribution years*
- **Identification**
 - *Flexirente effects post-2017*

- Rente mit 63 causally reduces employment
 - *by ~22 percentage points (DiD = -0.219^{***})*
- No uniform heterogeneity across subgroups
 - *region and education show no significant differential effects*
- Strong income gradient
 - *low earners (-35 pp) react most, top earners barely affected (-5 pp)*

Thank You!

for your attention. We look forward to your questions.

SLIDES & PAPER

